

JavaScript Events - Complete Guide

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What are Events?

Events are **actions performed by the user on the browser** or actions that happen in the browser environment. When any interaction occurs (like clicking, typing, scrolling), the browser creates an event object containing information about what happened.

- Actions Performed by user on browser are referred as events.
- Whenever event occurs browser creates an object which contains all info about the event and object on which event occurred.
- Ex : if user click on `<h1>` tag, browser automatically creates an object which has info about `<h1>` tag and type of event occurred (here, type is `click`).

Event Flow

User Action → Event Object Creation → Event Listener → Response Action

Event Object

The **Event Object** is automatically created by the browser when an event occurs. It contains:

- Event object is an object created by the browser when user performs some action, which holds all information about type of event occurred and the element on which the event occurred.
- Event object is passed to respective event handler, every time event occurs.
- So, we can access event object in callback function.
- **Type of event** (click, keydown, mouseover, etc.)

- **Target element** (which element triggered the event)
- **Additional properties** specific to the event type

```
// Event object example
button.addEventListener("click", function(e) {
  console.log(e.type);      // "click"
  console.log(e.target);    // The button element
  console.log(e.timeStamp); // When the event occurred
});
```

Event Listeners

Event Listeners are functions that wait for specific events to occur and execute code when those events happen.

Syntax

```
element.addEventListener(eventType, callbackFunction, options);
```

Ways to Handle Events

1. Inline Event Handlers (HTML)

```
<!-- Direct in HTML -->
<button onclick="changeText()">Click Me</button>
```

```
function changeText() {
  const h2 = document.querySelector("h2");
  h2.textContent = "Namaste World..!!";
}
```

Pros: Simple and direct **Cons:** Mixes HTML and JavaScript, harder to maintain

2. Element Property Assignment

```
const button = document.querySelector("button");
button.onclick = function() {
  const h2 = document.querySelector("h2");
  h2.textContent = "Namaste Duniya..!!";
};
```

Pros: Separates HTML and JavaScript **Cons:** Can only assign one function per event

3. `addEventListener()` Method (Recommended)

```
const button = document.querySelector("button");
button.addEventListener("click", function() {
  console.log("Clicked...!!!");
  console.log("this:", this); // refers to the button element
  const body = document.body;
  body.style.cssText = "color: white; background: black";
});
```

Pros:

- Can attach multiple listeners to same event
- More control and flexibility
- Better for complex applications

Mouse Events

Complete List of Mouse Events

Event	Description	When it fires
<code>click</code>	Single mouse click	On mouse button press and release
<code>dblclick</code>	Double mouse click	Two rapid clicks
<code>mousedown</code>	Mouse button pressed	When mouse button is pressed down
<code>mouseup</code>	Mouse button released	When mouse button is released
<code>mouseover</code>	Mouse enters element	When mouse pointer enters element
<code>mouseout</code>	Mouse leaves element	When mouse pointer leaves element
<code>mouseenter</code>	Mouse enters element	Similar to <code>mouseover</code> but doesn't bubble
<code>mouseleave</code>	Mouse leaves element	Similar to <code>mouseout</code> but doesn't bubble
<code>mousemove</code>	Mouse moves	When mouse pointer moves
<code>contextmenu</code>	Right mouse click	When right mouse button is clicked
<code>wheel</code>	Mouse wheel scroll	When mouse wheel is scrolled

Mouse Event Examples

Basic Click Event

```
const button = document.querySelector("button");
button.addEventListener("click", function() {
  console.log("Button clicked!");
});
```

Mouse Down/Up Events

```
const button = document.querySelector("button");
const body = document.body;
button.addEventListener("mousedown", function() {
  body.style.background = "red";
});
button.addEventListener("mouseup", function() {
  body.style.background = "white";
});
```

Mouse Hover Effects

```
button.addEventListener("mouseover", function() {
  const body = document.body;
  body.style.background = "black";
});
button.addEventListener("mouseout", function() {
  const body = document.body;
  body.style.background = "white";
});
```

Context Menu (Right Click)

```
const body = document.body;
body.addEventListener("contextmenu", function(e) {
  e.preventDefault(); // Prevents default context menu
  console.log("Right Clicked...");
});
```

Mouse Wheel Event

```
body.addEventListener("wheel", function(e) {
  console.log("Scroll direction:", e.deltaY > 0 ? "Down" : "Up");
  console.log("Scroll amount:", e.deltaY);
});
```

Keyboard Events

Types of Keyboard Events

Event	Description	When it fires
keydown	Key is pressed down	When any key is pressed (repeats if held)
keyup	Key is released	When any key is released
keypress	Key is pressed	Deprecated - use keydown instead

Keyboard Event Properties

```

window.addEventListener("keydown", function(e) {
  console.log("Key:", e.key);           // The key value (e.g., "a", "Enter", "Shift")
  console.log("Code:", e.code);         // Physical key code (e.g., "KeyA", "Enter")
  console.log("Ctrl:", e.ctrlKey);      // Boolean - was Ctrl pressed?
  console.log("Shift:", e.shiftKey);    // Boolean - was Shift pressed?
  console.log("Alt:", e.altKey);        // Boolean - was Alt pressed?
  console.log("Meta:", e.metaKey);      // Boolean - was Meta/Cmd pressed?
});

```

Keyboard Event Examples

Basic Key Detection

```

window.addEventListener("keydown", function(e) {
  console.log("Key pressed:", e.key);

  if (e.key === "Enter") {
    console.log("Enter key was pressed!");
  }
});

```

Keyboard Shortcuts (Fixed Version)

```

// Dark mode toggle with Ctrl+D
window.addEventListener("keydown", function(e) {
  if (e.ctrlKey && e.key === "d") {
    e.preventDefault(); // Prevent browser's default bookmark action

    const body = document.body;
    // Toggle between light and dark mode
    if (body.style.backgroundColor === "black") {
      body.style.backgroundColor = "white";
      body.style.color = "black";
    } else {
      body.style.backgroundColor = "black";
      body.style.color = "white";
    }
  }
});

```

Alert on Specific Key

```

window.addEventListener("keydown", function(e) {
  if (e.key === "a" || e.key === "A") {
    e.preventDefault();
    alert("Welcome to another World");
  }
});

```

Form Events

Common Form Events

Event	Description	When it fires
<code>submit</code>	Form submission	When form is submitted
<code>reset</code>	Form reset	When form is reset
<code>focus</code>	Element gains focus	When element becomes active
<code>blur</code>	Element loses focus	When element becomes inactive
<code>change</code>	Value changes	When input value changes and loses focus
<code>input</code>	Value being typed	Real-time as user types

Form Event Examples

Form Submission Handling

```
const form = document.querySelector("form");

form.addEventListener("submit", function(e) {
  e.preventDefault(); // Prevent default form submission

  // Get form data
  const formData = new FormData(this);
  const username = formData.get("username");
  const email = formData.get("emailid");
  const password = formData.get("password");

  console.log("Username:", username);
  console.log("Email:", email);
  console.log("Password:", password);

  // Validate data here
  if (!username || !email || !password) {
    alert("Please fill all required fields");
    return;
  }

  alert("Form submitted successfully!");
});
```

Input Validation

```
const usernameInput = document.querySelector("#inp1");
const emailInput = document.querySelector("#inp2");

// Real-time validation
usernameInput.addEventListener("input", function(e) {
  const value = e.target.value;
  if (value.length < 3) {
    e.target.style.borderColor = "red";
  } else {
    e.target.style.borderColor = "green";
  }
});

// Email validation on blur
emailInput.addEventListener("blur", function(e) {
  const email = e.target.value;
  const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

  if (!emailRegex.test(email)) {
    e.target.style.borderColor = "red";
    alert("Please enter a valid email address");
  } else {
    e.target.style.borderColor = "green";
  }
});
```

Event Methods

preventDefault()

Stops the default action of an event from happening.

```
// Prevent form submission
form.addEventListener("submit", function(e) {
  e.preventDefault();
  console.log("Form submission prevented");
});

// Prevent right-click context menu
document.addEventListener("contextmenu", function(e) {
  e.preventDefault();
});
```

stopPropagation()

Stops the event from bubbling up to parent elements.

```
button.addEventListener("click", function(e) {
  e.stopPropagation();
  console.log("Event stopped here");
});
```

removeEventListener()

Removes a previously added event listener.

```
function clickHandler() {
  console.log("Clicked!");
}

// Add listener
button.addEventListener("click", clickHandler);

// Remove listener
button.removeEventListener("click", clickHandler);
```

Advanced Event Concepts

Event Bubbling and Capturing

Events in JavaScript follow a specific flow:

1. **Capturing Phase:** Event travels from document to target element
2. **Target Phase:** Event reaches the target element
3. **Bubbling Phase:** Event bubbles back up to document

```
// Capturing (third parameter = true)
element.addEventListener("click", handler, true);

// Bubbling (default)
element.addEventListener("click", handler, false);
```

Event Delegation

Handle events for multiple elements using a single listener on a parent element.

```
// Instead of adding listeners to each button
const container = document.querySelector(".button-container");

container.addEventListener("click", function(e) {
  if (e.target.tagName === "BUTTON") {
    console.log("Button clicked:", e.target.textContent);
  }
});
```

Complete Working Examples

1. Theme Switcher (Corrected Version)

```
// Better implementation with error handling
const lightBtn = document.querySelector("#light");
const darkBtn = document.querySelector("#dark");

function switchTheme(theme) {
  const link = document.querySelector("link[rel='stylesheet']");
  if (link) {
    link.href = `${theme}.css`;
  } else {
    console.error("Stylesheet link not found");
  }
}

if (lightBtn) {
  lightBtn.addEventListener("click", function() {
    switchTheme("light");
  });
}

if (darkBtn) {
  darkBtn.addEventListener("click", function() {
    switchTheme("dark");
  });
}
```

2. Advanced Keyboard Controls

```
// Improved keyboard event handling
const keyStates = {
  ctrl: false, shift: false, alt: false
};

window.addEventListener("keydown", function(e) {
  // Track modifier keys
  keyStates.ctrl = e.ctrlKey;
  keyStates.shift = e.shiftKey;
  keyStates.alt = e.altKey;

  console.log(`Key pressed: ${e.key}`);

  // Keyboard shortcuts
  if (e.ctrlKey && e.key === "d") {
    e.preventDefault();
    toggleDarkMode();
  }
  if (e.ctrlKey && e.key === "s") {
    e.preventDefault();
    saveContent();
  }
  // Special key handling
  if (e.key === "Escape") {
    closeModal();
  }
  if (e.key === "Enter" && e.ctrlKey) {
    submitForm();
  }
});

window.addEventListener("keyup", function(e) {
  // Update modifier key states
  keyStates.ctrl = e.ctrlKey;
  keyStates.shift = e.shiftKey;
  keyStates.alt = e.altKey;
});

function toggleDarkMode() {
  const body = document.body;
  const isDark = body.style.backgroundColor === "black";

  body.style.backgroundColor = isDark ? "white" : "black";
  body.style.color = isDark ? "black" : "white";
}

function saveContent() {
  console.log("Save shortcut triggered!");
  // Implement save functionality
}

function closeModal() {
  console.log("Escape pressed - closing modal");
  // Implement modal close
}

function submitForm() {
  console.log("Ctrl+Enter pressed - submitting form");
  // Implement form submission
}
```

Event Types Reference

Mouse Events Deep Dive

```
const element = document.querySelector("#target");

// Click events
element.addEventListener("click", (e) => {
  console.log("Single click");
});

element.addEventListener("dblclick", (e) => {
  console.log("Double click");
});

// Mouse button events
element.addEventListener("mousedown", (e) => {
  console.log(`Mouse button ${e.button} pressed`);
  // 0 = left, 1 = middle, 2 = right
});

element.addEventListener("mouseup", (e) => {
  console.log(`Mouse button ${e.button} released`);
});

// Mouse movement events
element.addEventListener("mouseover", (e) => {
  console.log("Mouse entered element");
  e.target.style.backgroundColor = "lightblue";
});

element.addEventListener("mouseout", (e) => {
  console.log("Mouse left element");
  e.target.style.backgroundColor = "";
});

// Mouse position tracking
element.addEventListener("mousemove", (e) => {
  console.log(`Mouse position: (${e.clientX}, ${e.clientY})`);
});
```

Keyboard Events Deep Dive

```
// Key detection with modifiers
window.addEventListener("keydown", function(e) {
  const keyCombo = [];
  if (e.ctrlKey) keyCombo.push("Ctrl");
  if (e.shiftKey) keyCombo.push("Shift");
  if (e.altKey) keyCombo.push("Alt");
  if (e.metaKey) keyCombo.push("Meta");

  keyCombo.push(e.key);

  console.log("Key combination:", keyCombo.join(" + "));

  // Handle special combinations
  switch(true) {
    case e.ctrlKey && e.key === "z":
      e.preventDefault();
      console.log("Undo action");
      break;

    case e.ctrlKey && e.key === "y":
      e.preventDefault();
      console.log("Redo action");
      break;

    case e.ctrlKey && e.shiftKey && e.key === "N":
      e.preventDefault();
      console.log("New private window");
      break;
  }
});
```

Window and Document Events

```
// Page load events
window.addEventListener("load", function() {
  console.log("Page fully loaded");
});
document.addEventListener("DOMContentLoaded", function() {
  console.log("DOM ready");
});

// Window resize
window.addEventListener("resize", function() {
  console.log(`Window size: ${window.innerWidth}x${window.innerHeight}`);
});

// Scroll events
window.addEventListener("scroll", function() {
  console.log(`Scroll position: ${window.scrollY}`);
});

// Before page unload
window.addEventListener("beforeunload", function(e) {
  e.preventDefault();
  e.returnValue = ""; // Shows confirmation dialog
});
```

Best Practices

1. Always Use addEventListener()

- More flexible than other methods
- Allows multiple listeners per event
- Better for maintainable code

2. Use Event Delegation for Dynamic Content

```
// Good for dynamically added elements
document.addEventListener("click", function(e) {
  if (e.target.classList.contains("dynamic-button")) {
    handleDynamicClick(e);
  }
});
```

3. Clean Up Event Listeners

```
// Store reference to remove later
const clickHandler = function(e) {
  console.log("Clicked!");
};

element.addEventListener("click", clickHandler);

// Remove when no longer needed
element.removeEventListener("click", clickHandler);
```

4. Use Debouncing for Expensive Operations

```
function debounce(func, wait) {
  let timeout;
  return function executedFunction(...args) {
    const later = () => {
      clearTimeout(timeout);
      func(...args);
    };
    clearTimeout(timeout);
    timeout = setTimeout(later, wait);
  };
}

// Debounced scroll handler
const debouncedScroll = debounce(function() {
  console.log("Scroll handled");
}, 100);

window.addEventListener("scroll", debouncedScroll);
```

5. Error Handling

```
element.addEventListener("click", function(e) {  
  try {  
    // Your event handling code  
    performAction();  
  } catch (error) {  
    console.error("Event handler error:", error);  
  }  
});
```

Common Errors and Fixes

Error 1: Event Listener Not Working

Problem: Element not found when script runs

```
// ❌ Bad - script runs before DOM is ready  
const button = document.querySelector("button");  
button.addEventListener("click", handler); // button might be null
```

Solution:

```
// ✅ Good - wait for DOM or check if element exists  
document.addEventListener("DOMContentLoaded", function() {  
  const button = document.querySelector("button");  
  if (button) {  
    button.addEventListener("click", handler);  
  }  
});
```

Error 2: this Context Confusion

```
// ❌ Arrow functions don't have their own 'this'  
button.addEventListener("click", (e) => {  
  console.log(this); // Window object, not button  
});  
  
// ✅ Regular function maintains 'this' context  
button.addEventListener("click", function(e) {  
  console.log(this); // Button element  
});
```

Error 3: Memory Leaks

```
// ✗ Bad - creates new function each time
for (let i = 0; i < buttons.length; i++) {
  buttons[i].addEventListener("click", function() {
    console.log(`Button ${i} clicked`);
  });
}

// ☑ Good - reuse function or use event delegation
function buttonClickHandler(e) {
  console.log("Button clicked:", e.target.textContent);
}
buttons.forEach(button => {
  button.addEventListener("click", buttonClickHandler);
});
```

Performance Tips

1. Passive Event Listeners

```
// For scroll and touch events
window.addEventListener("scroll", handler, { passive: true });
```

2. Once Option

```
// Listener runs only once then removes itself
button.addEventListener("click", handler, { once: true });
```

3. Throttling for High-Frequency Events

```
function throttle(func, limit) {
  let inThrottle;
  return function() {
    const args = arguments;
    const context = this;
    if (!inThrottle) {
      func.apply(context, args);
      inThrottle = true;
      setTimeout(() => inThrottle = false, limit);
    }
  };
}

// Throttled mousemove
element.addEventListener("mousemove", throttle(function(e) {
  console.log("Mouse moved");
}, 100));
```

Modern Event Handling Patterns

1. Event Handler Classes

```
class EventManager {
  constructor() {
    this.handlers = new Map();
    this.init();
  }
  init() {
    this.bindEvents();
  }
  bindEvents() {
    document.addEventListener("click", this.handleClick.bind(this));
    document.addEventListener("keydown", this.handleKeyDown.bind(this));
  }

  handleClick(e) {
    if (e.target.matches("[data-action]")) {
      const action = e.target.dataset.action;
      this.executeAction(action, e);
    }
  }
  handleKeyDown(e) {
    const key = `${e.ctrlKey ? "ctrl+" : ""}${e.key}`;
    if (this.handlers.has(key)) {
      e.preventDefault();
      this.handlers.get(key)(e);
    }
  }

  registerShortcut(key, handler) {
    this.handlers.set(key, handler);
  }

  executeAction(action, event) {
    switch(action) {
      case "toggle-theme":
        this.toggleTheme();
        break;
      case "save":
        this.save();
        break;
      default:
        console.log(`Unknown action: ${action}`);
    }
  }

  toggleTheme() {
    document.body.classList.toggle("dark-theme");
  }

  save() {
    console.log("Saving...");
  }
}

// Usage
const eventManager = new EventManager();
eventManager.registerShortcut("ctrl+s", () => console.log("Save shortcut"));
eventManager.registerShortcut("ctrl+d", () => console.log("Dark mode shortcut"));
```


2. Custom Events

```
// Create custom events
const customEvent = new CustomEvent("themeChanged", {
  detail: { theme: "dark", timestamp: Date.now() }
});

// Listen for custom events
document.addEventListener("themeChanged", function(e) {
  console.log("Theme changed to:", e.detail.theme);
});

// Dispatch custom event
document.dispatchEvent(customEvent);
```

Browser Compatibility Notes

- `addEventListener()` is supported in all modern browsers
- Some older browsers might need polyfills for newer event types
- Always test keyboard shortcuts across different operating systems
- Mobile devices have touch events instead of mouse events

Summary

Events are the foundation of interactive web applications. Key takeaways:

1. Always use `addEventListener()` for maximum flexibility
2. Handle errors gracefully with try-catch blocks
3. Prevent default actions when needed with `e.preventDefault()`
4. Clean up event listeners to prevent memory leaks
5. Use event delegation for dynamic content
6. Test thoroughly across different browsers and devices

